Skills Summary

Percent Applications: Tax, Tip, Discount, and Markup

Use this reference while completing your worksheet or when studying.

1. Sales tax — a percent added to the price

Name the base first: tax is charged *on the price*, so the price is the base (100%).

$$\text{total} = \text{price} \times \left(1 + \frac{\text{tax rate}}{100}\right) \qquad \text{tax} = \text{price} \times \frac{\text{tax rate}}{100}$$

A 8% tax has multiplier 1.08: the 1 keeps the price, the 0.08 adds the tax. Read the question — "how much tax" wants the second formula, "how much in all" wants the first.

Example: A lamp costs \$52 and the tax rate is 8%. Find the total paid, in dollars.
Step 1. The question asks for the amount paid in all, and tax is added to the price, so use the price as the base and the add-on multiplier.
Step 2. Multiplier: $1 + \frac{8}{100} = 1.08$; $52 \cdot 1.08 = 56.16$.
Step 3. Total paid: **56.16** dollars.

Try it: A \$38 shirt is taxed at 6%. Find the total paid, in dollars.

check: **40.28**

2. Tip — a percent added to the bill

The base is the bill before the tip. A tip works exactly like tax:

$$\text{amount paid} = \text{bill} \times \left(1 + \frac{\text{tip percent}}{100}\right)$$

Mental math: 10% of a bill is the bill with the decimal point moved one place left; 5% is half of that; 20% is double it.

Example: A bill is \$46 and the tip is 20%. Find the amount paid, in dollars.
Step 1. The tip is added to the bill, so the bill is the base and the amount paid uses the add-on multiplier.
Step 2. Multiplier: $1 + \frac{20}{100} = 1.20$; $46 \cdot 1.20 = 55.2$.
Step 3. Amount paid: **55.20** dollars.

Try it: A \$62 bill with a 15% tip. Find the amount paid, in dollars.

check: **71.30**

3. Discount — a percent taken off

Subtract inside the percent, not at the end. If 30% comes off, the customer still pays $100 - 30 = 70\%$:

$$\text{sale price} = \text{original} \times \frac{100 - \text{percent off}}{100}$$

Backwards: when the sale price is given, the original price is the unknown base — put x where "original" stands above and solve for x.

Example: A \$84 coat is 30% off. Find the sale price, in dollars.

Step 1. The percent comes off the original price, so the original is the base and the customer pays the remaining percent.

Step 2. Remaining: $100 - 30 = 70$, so multiply by $\frac{70}{100} = 0.7$; $84 \cdot 0.7 = 58.8$.

Step 3. Sale price: **58.80** dollars.

Try it: A \$50 game is 35% off. Find the sale price, in dollars.

check: **32.50**

4. Markup — a percent added to a store's cost

The base is what the store paid, not what the customer pays:

$$\text{selling price} = \text{cost} \times \left(1 + \frac{\text{markup percent}}{100}\right)$$

Markup and tax use the same multiplier shape. Only the base changes: markup sits on the store's cost, tax sits on the shelf price.

Example: A store pays \$24 for a kite and marks it up 40%. Find the selling price, in dollars.

Step 1. Markup is added to what the store paid, so the cost is the base and the multiplier adds the markup on.

Step 2. Multiplier: $1 + \frac{40}{100} = 1.40$; $24 \cdot 1.40 = 33.6$.

Step 3. Selling price: **33.60** dollars.

Try it: A store pays \$65 for a jacket and marks it up 25%. Find the selling price, in dollars.

check: **81.25**

(!) Watch out: Two percents in one problem act on *different* bases. Take 25% off and then add 6% tax by multiplying twice ($\times 0.75$, then $\times 1.06$). Folding them into one 19% change is the most common wrong answer on this topic.